6.10.4 Wrap-Up: Reflection

1. One thing I struggled with today was …

2. One thing I need to practice more is …

3. One thing I understand very well is …

Unit 6, Lesson 11: Solving Problems Using Similarity – Part 1

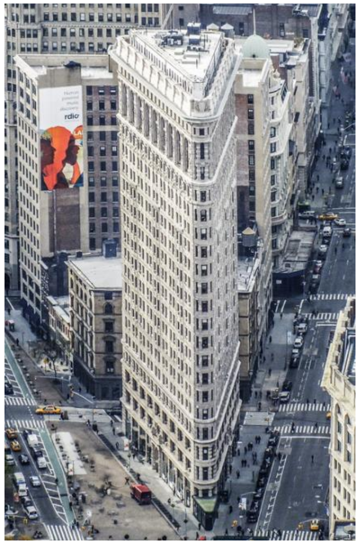
Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

Learning Target

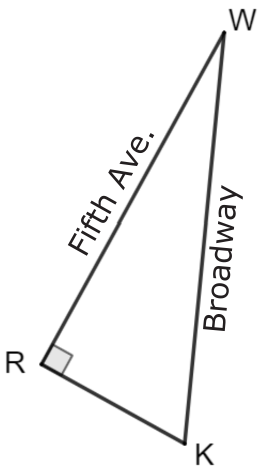
☐ I can solve mathematical problems involving similarity in two-dimensional figures.

6.11.1 Warm-Up: The Flatiron Building



The Flatiron Building, designed by Daniel Burnham and built in 1902, is a triangular 22-story building located in New York City. The building sits on a triangular block formed by 5th Avenue, Broadway, and East 22nd Street. The building got its name from an old-fashioned flat iron.

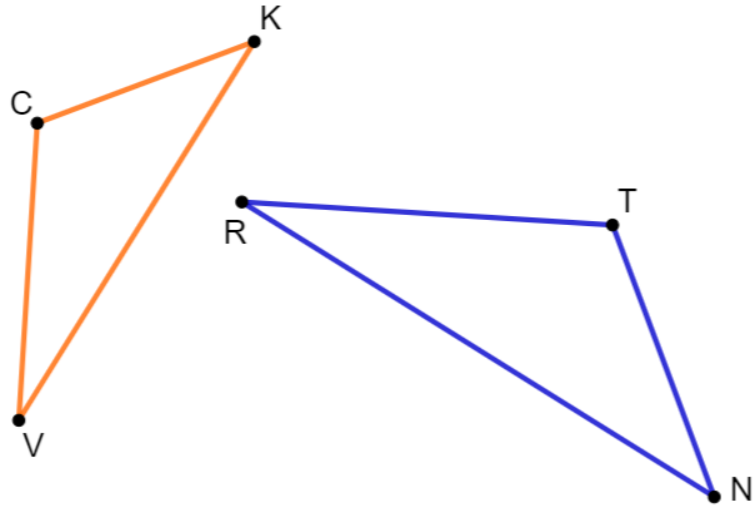
Although the ground base appears to be an isosceles triangle, the base of the Flatiron Building is actually a right triangle. Morgan used a map to draw a triangle that models the base of the building.



1. What other measurement should Morgan check to be sure her drawing is similar to the building?

6.11.2 Collaboration: Similar Figures

1. ΔCKV and ΔTNR are shown, where $\Delta CKV \sim \Delta TNR$.



   a. Some measurements of the angles in ΔCKV and ΔTNR are given.
   $m\angle TRN = 28°$, $m\angle VCK = (8x + 26)°$, and
   $m\angle RTN = (12x - 8)°$.

   Determine each.

   $x =$ ____________ $m\angle CKV =$ ____________

Some measurements of the sides of ΔCKV and ΔTNR are given.

$VK = 8$, $CK = 5z + 1.7$, $CV = 5.4$, $TR = 2y - 1.25$, $RN = 10$, and $TN = 5.25$.

b. What is the scale factor from ΔCKV to ΔTNR?

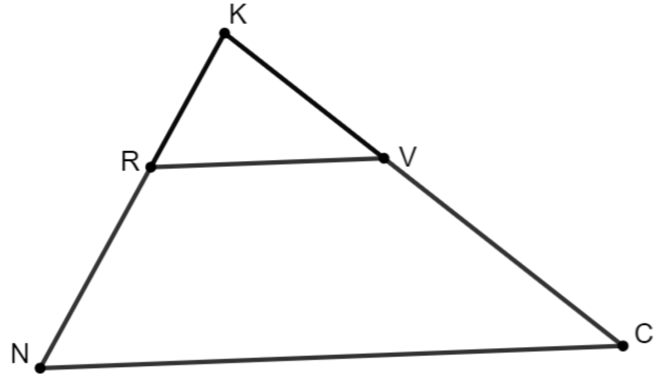
c. Determine the value of each variable.

$z =$____________ $y =$____________

d. Determine the measure of each side.

$CK =$____________ $TR =$____________

2. ΔKVR and ΔKCN are shown, where $\Delta KVR \sim \Delta KCN$.



Some measurements of the sides are given.
$RV = 4.6$, $KR = 3$, $KV = 5x - 11$, $NC = 11.5$, and $KC = 3x + 1$.

Maria used the information to determine the length of $\overline{KC}$. Her work is shown.

Maria's Work	
First, Maria found the scale factor.	$\frac{NC}{RV} = \frac{11.5}{4.6} = 2.5$
Second, she used the scale factor to write an equation. Then she solved for x.	$5x - 11 = 2.5(3x + 1)$ $5x - 11 = 7.5x + 2.5$ $-2.5x = -13.5$ $x = -5.4$
Lastly, she substituted x into an equation.	$KC = 3(-5.4) + 1$ $KC = -15.2$

Maria knows side lengths cannot be negative values, but she does not know what she did wrong.

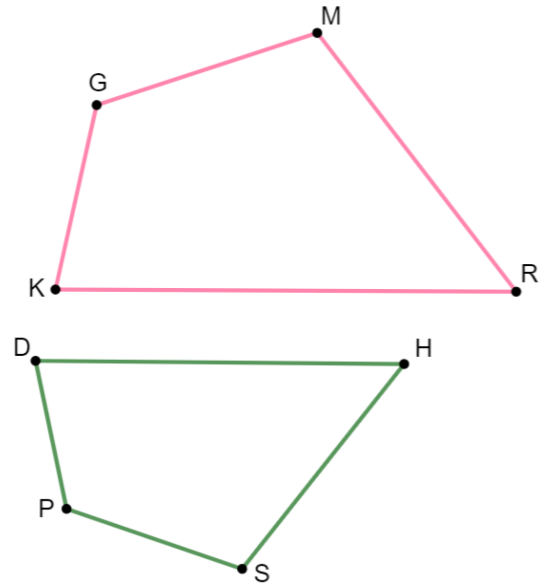
a. Find and circle Maria's error.

b. Write a note to Maria explaining what her mistake was and how she can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

6.11.3 Your Turn!

1. $GMRK$ and $PSHD$ are shown, where $GMRK \sim PSHD$.



Some measurements of the sides of $GMRK$ and $PSHD$ are given.

$KG = 12$, $GM = 4x - 12$, $MR = 7k - 15$, $DP = 9.6$, $SP = 12.8$, $KR = 30$, and $SH = 4k - 4$.

a. Determine the value of each variable.

$x =$__________ $k =$__________

b. Determine the value of each side.

$GM =$ ____________ $DH =$ ____________

$SH =$ ____________

c. What is the perimeter of $GMRK$?

Some measurements of the angles of $GMRK$ and $PSHD$ are given.

$m\angle KGM = (11w - 12)°$, $m\angle GMR = 110°$,
$m\angle GKR = (6z + 18)°$, $m\angle DPS = (8w + 24)°$, and
$m\angle PDH = (8z - 2)°$.

d. Determine the value of each variable.

$w =$ ____________ $z =$ ____________

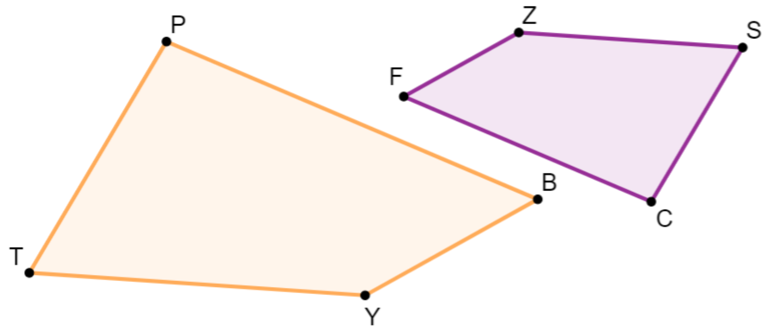
e. Determine the value of each angle.

$m\angle KGM =$ ________ $m\angle DHS =$ ________

$m\angle PDH =$ ________

6.11.4 Wrap-Up: Error Analysis

1. Cheryl and Austin are finding the measure of side SC. They were given the following information about similar quadrilaterals $ZSCF$ and $YTPB$ that are shown.
$\Delta ZSCF \sim \Delta YTPB, ZS = 15$, $SC = 5x - 8$, $PT = 3x + 6$, and $YT = 22.5$.



Cheryl's and Austin's equations are shown.

Cheryl's Work	Austin's Work
$15(5x - 8) = 22.5(3x + 6)$	$1.5(5x - 8) = 3x + 6$

a. Determine whose equation, if any, is set up incorrectly. Show your work or write an explanation.

b. What is the length of $\overline{SC}$?

Unit 6, Lesson 12: Solving Problems Using Similarity – Part 2

Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

Learning Target

☐ I can solve real-world problems involving similarity in two-dimensional figures.